

DATA ANALYST

PIZZA SALES ANALYSIS



PROBLEM STATEMENT

The objective of this project is to analyze pizza sales data to understand revenue trends, best-selling pizzas, customer preferences, and overall business performance, and to provide actionable insights to improve sales and profitability.

DATASET DESCRIPTION

The dataset consists of four tables that represent a pizza sales system:

1. orders:

- order_id: Unique identifier for each order
- order_date: Date when the order was placed
- order_time: Time when the order was placed

2. order_details:

- order_details_id: Unique identifier for each order item
- order_id: Reference to the orders table
- pizza_id: Identifier of the pizza ordered
- quantity: Number of pizzas ordered

3. pizzas:

- pizza_id: Unique identifier for each pizza (includes size)
- pizza_type_id: Reference to pizza type
- size: Size of the pizza (S, M, L)
- price: Price of the pizza

4. pizza_types:

- pizza_type_id: Unique identifier for pizza category
- name: Name of the pizza
- category: Type of pizza (Chicken, Veg, etc.)

- ingredients: Ingredients used in the pizza

These tables are connected using primary and foreign key relationships to perform comprehensive sales analysis.

TOOLS USED

- **SQL (for data analysis)**

- **Excel / Power BI (for visualization – optional)**

ANALYSIS QUERIES

-- retrieve the total number of orders placed

USE pizzahut;

SELECT

 COUNT(order_id) AS total_orders

FROM

 orders;

-- calculate the total revenue generated from pizza sales.

SELECT

 ROUND(SUM(order_details.quantity * pizzas.price),

 2) AS total_sales

FROM

 order_details

 JOIN

pizzas ON pizzas.pizza_id = order_details.pizza_id;

-- identify the highest priced pizza

SELECT

 pizza_types.name, pizzas.price

FROM

 pizza_types

 JOIN

 pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id

ORDER BY pizzas.price DESC

LIMIT 1;

-- identify the most common pizza size ordered

SELECT

 pizzas.size,

 COUNT(order_details.order_details_id) AS order_count

FROM

 pizzas

 JOIN

 order_details ON pizzas.pizza_id = order_details.pizza_id

GROUP BY pizzas.size

ORDER BY order_count DESC;

-- list the top 5 most ordered pizza types along with their quantities.

SELECT

 pizza_types.name, SUM(order_details.quantity) AS quantity

FROM

 pizza_types

 JOIN

 pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id

 JOIN

 order_details ON order_details.pizza_id = pizzas.pizza_id

GROUP BY pizza_types.name

ORDER BY quantity DESC

LIMIT 5;

-- join the necessary tables to find the total quantity of each pizza category ordered.

SELECT

 pizza_types.category,

 SUM(order_details.quantity) AS quantity

FROM

 pizza_types

 JOIN

 pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id

 JOIN

```
order_details ON order_details.pizza_id = pizzas.pizza_id
```

```
GROUP BY pizza_types.category
```

```
ORDER BY quantity DESC;
```

```
-- determine the distribution of orders by hour of the day.
```

```
SELECT
```

```
    HOUR(order_time), COUNT(order_id) AS order_count
```

```
FROM
```

```
    orders
```

```
GROUP BY HOUR(order_time);
```

```
-- join relevant tables to find the category-wise distribution of pizzas.
```

```
SELECT
```

```
    category, COUNT(name)
```

```
FROM
```

```
    pizza_types
```

```
GROUP BY category;
```

```
-- group the orders by date and calculate the average number of pizzas ordered per day.
```

```
SELECT
```

```
    ROUND(AVG(quantity), 0) as avg_pizza_ordered_per_day
```

```
FROM
```

```
    (SELECT
```

```
orders.order_date, SUM(order_details.quantity) AS quantity  
FROM  
orders  
JOIN order_details ON orders.order_id = order_details.order_id  
GROUP BY orders.order_date) AS order_quantity;  
  
-- determine the top 3 most ordered pizza types based on revenue.
```

```
SELECT  
pizza_types.name,  
SUM(order_details.quantity * pizzas.price) AS revenue  
FROM  
pizza_types  
JOIN  
pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id  
JOIN  
order_details ON order_details.pizza_id = pizzas.pizza_id  
GROUP BY pizza_types.name  
ORDER BY revenue DESC  
LIMIT 3;  
  
-- calculate the percentage contribution of each pizza type to total revenue  
  
SELECT  
pizza_types.category,
```

```
ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
ROUND(SUM(order_details.quantity * pizzas.price),
2) AS total_sales
FROM
order_details
JOIN
pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100, 2)as revenue
FROM
pizza_types
JOIN
pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN
order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

-- analyze the cumulative revenue generated over time

```
select order_date,
sum(revenue) over(order by order_date) as cum_revenue
from
(select orders.order_date,
```

```
sum(order_details.quantity * pizzas.price) as revenue
```

```
from order_details join pizzas
```

```
on order_details.pizza_id = pizzas.pizza_id
```

```
join orders
```

```
on orders.order_id = order_details.order_id
```

```
group by orders.order_date )as sales;
```

-- determine the top 3 most ordered pizza types based on revenue for each pizza category

```
select name, revenue from
```

```
(select category,name, revenue,
```

```
rank() over(partition by category order by revenue desc) as rn
```

```
from
```

```
(select pizza_types.category, pizza_types.name,
```

```
sum(order_details.quantity * pizzas.price) as revenue
```

```
from pizza_types join pizzas
```

```
on pizza_types.pizza_type_id = pizzas.pizza_type_id
```

```
join order_details
```

```
on order_details.pizza_id = pizzas.pizza_id
```

```
group by pizza_types.category, pizza_types.name) as a) as b
```

```
where rn <= 3
```

KEY INSIGHTS

- Certain pizza categories generate significantly higher revenue than others.
- A few top-selling pizzas contribute to a large portion of total sales.
- Sales show variation across different months, indicating seasonal trends.
- Larger pizza sizes contribute more to overall revenue.
- High Average Order Value suggests customers prefer multiple or premium items.
- Classic category pizzas are the best-selling, indicating strong customer preference for traditional options.
- Revenue tends to increase towards the end of the year, suggesting seasonal demand patterns or higher customer spending during that period.
- Peak order time is around 12 PM, indicating that lunch hours drive the highest pizza sales.
- Customer demand is time-sensitive, with a clear spike during midday hours.

BUSINESS RECOMMENDATIONS

- Focus marketing on top-performing pizzas to maximize revenue.
- Introduce discounts or combo offers for low-selling pizzas.

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- **Promote larger size pizzas to increase average order value.**
 - **Analyze seasonal trends to plan inventory and promotions.**
 - **Bundle popular items to boost overall sales.**